

EE 412/512 HW Assignment #1

Due September 28, 2025

Write the following statement on the first line of your solution sheet:

“I completed this homework assignment independently.”

Register and download ADS using Bilkent e-mail and install it on your PC. Read the ADS Tutorial.

- 1) Design a third-order Chebychev (0.2 dB) LC-bandpass filter with a center frequency of f_0 and a fractional bandwidth of Δf . Choose f_0 randomly between 50 MHz and 100 MHz and $\Delta f / f_0$ between 0.1 and 0.2. The source and load impedances are both 50Ω . Refer to the accompanying document for the filter design procedure (refer to FilterDesign.pdf).
 - a) Verify your filter design using ADS. Plot S_{11} and S_{21} in dB as a function of frequency (from $f_0 - 2\Delta f$ to $f_0 + 2\Delta f$). Note that for a good 0.2dB Chebychev filter, S_{11} should be less than -12dB , and S_{21} should be no worse than -0.2dB within the band.
 - b) Assume that the inductors have $Q=80$. Simulate your filter and plot the S_{11} and S_{21} response.
 - c) Assume that there is a parasitic capacitance of 100 fF from *every* node to the ground. Find out and plot how the S_{11} and S_{21} responses are degraded because of parasitic capacitors.
 - d) Transform your filter structure so that there are no series LC branches. With all nodes containing a shunt capacitor, the parasitic capacitor of 100 fF can be used as a part of the shunt capacitance. Plot the S_{11} and S_{21} response of the modified filter. Hint: Refer to FilterDesign.pdf.
- 2) A 0.15λ long transmission line of characteristic impedance $Z_0=50\Omega$ is terminated with a load impedance of $Z_L=R+jX$ at 3 GHz. (Choose random R and X values between 10Ω and 250Ω)
 - a) Find the impedance at the other end of the line using the relevant formula.
 - b) Find the value of the input impedance using the Smith Chart. Show your work on the Smith chart (can be downloaded from Moodle).
 - c) Verify your result in a) and b) using ADS with $f=3$ GHz. You may use a transmission line element with an electrical length of 54 degrees at 3 GHz.
 - d) Find the approximate transmission line length in λ units (instead of 0.15λ) to make the input impedance purely real using the Smith chart. What are the possible values of the real impedance? Show your work on the Smith chart.
 - e) Verify your result in d) using ADS at $f=3$ GHz. Note that you can use the real and imaginary parts of the Z_{11} of the network to find the input impedance.